

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Pharmacy SEMESTER - 8 Winter 2025 (Supply.)

Course :Bachelor of Pharmacy Branch : Pharmacy

Semester : SEMESTER - 8

Subject Code & Name: ET - 1 - BP814ET - PHARMACEUTICAL PRODUCT DEVELOPMENT

Time : 3 Hours]

[Total Marks : 75

Instructions to the Students:

- 1.All questions are compulsory.
- 2.Draw diagrams / figures wherever necessary
- 3.Figures to right indicate full marks

Q.1 Objective Type Questions (Answer all the questions) (10 x 2) = 20

- | | |
|---|---|
| A) Define a directly compressible vehicle with an example. | 2 |
| B) Write about the LAL test. | 2 |
| C) Define propellants used in aerosols with examples | 2 |
| D) What are suspending agents? Give Examples | 2 |
| E) What are surfactants? Classify them. | 2 |
| F) What are coating materials? Explain various types of coating. | 2 |
| G) Differentiate primary, secondary and tertiary packaging systems. | 2 |
| H) Define leaching with a suitable example. | 2 |
| I) Mention the ideal qualities of closures. | 2 |
| J) Write about the accelerated stability study protocol of the solid dosage form as per ICH guidelines. | 2 |

Q.2 Long Answers (Answer 2 out of 3) (2 x 10) = 20

- | | |
|---|----|
| A) What is QbD? Describe advantages and challenges of QbD approach in product development. | 10 |
| B) Describe the various techniques used to improve the solubility of poorly soluble drugs. | 10 |
| C) Describe various excipients used in the formulation of NDDS. Mention their properties, merits and demerits | 10 |

Q.3 Short Answers (Answer 7 out of 9) (7 x 5) = 35

- | | |
|--|---|
| A) Define pharmaceutical product. Highlight the objectives of product development | 5 |
| B) Discuss general excipients used in parenteral formulations. | 5 |
| C) How to determine the compatibility of drugs and other additives used in the formulation of a dosage form. | 5 |
| D) Define the term 'factorial design.' Give the advantages and disadvantages of the same. | 5 |
| E) Write a short note on cyclodextrins and their applications. | 5 |
| F) Write about the glass packaging material for pharmaceutical product development. | 5 |
| G) Define and classify optimization techniques. | 5 |
| H) Write a short note on excipients in aerosol products. | 5 |
| I) Enumerate different types of solublizers with suitable examples. | 5 |

*** * * End * * ***